

Def. A linear operator $E: V \rightarrow V$ is called a projection if $E^2 = E$.

Lemma. Let $E: V \rightarrow V$ be a projection. Then,

1) $\text{Im } E = \{x \in V \mid E(x) = x\}$,

2) $V = \text{Im } E \oplus \ker E$

Proof. $x \in \text{Im } E \Leftrightarrow x = E(\alpha)$ for some $\alpha \in V \Rightarrow E(x) = E^2(\alpha) = E(\alpha) = x$. The converse is trivial!

Given $x \in V$, $x = E(x) + (x - E(x))$. Clearly $E(x) \in \text{Im } E$ and

$$E(x - E(x)) = E(x) - E^2(x) = (E - E^2)(x) = 0, \text{ so } x - E(x) \in \ker E.$$

And, $E(y) = y$ and $E(y) = 0$, then $y = 0$. Thus $\text{Im } E \cap \ker E = \{0\}$.

Thm. Let $V = W_1 \oplus W_2$. Then, there is one and only one projection $E: V \rightarrow V$ s.t.

$$\text{Im } E = W_1 \text{ and } \ker E = W_2$$

Proof. Let E be a projection. For β_1 , basis for $\text{Im } E$, and β_2 , basis for $\ker E$,

$$[E]_{\beta_1 \cup \beta_2} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ where } r = \text{rank } E.$$

Comment: In General,

let $\{\alpha_1, \dots, \alpha_k\}$ be a basis for $\text{Im } E$ and $\{\alpha_{k+1}, \dots, \alpha_n\}$ be a basis for $\ker E$.

Then, the invertible matrix $A = \begin{bmatrix} \alpha_1 & \alpha_2 & \dots & \alpha_n \end{bmatrix}$ satisfies

$$EA = \begin{bmatrix} E\alpha_1 & \dots & E\alpha_n \end{bmatrix} = \begin{bmatrix} \alpha_1 & \dots & \alpha_k & 0 & \dots & 0 \end{bmatrix}$$

Therefore, $E = \begin{bmatrix} \alpha_1 & \dots & \alpha_k & 0 & \dots & 0 \end{bmatrix} \cdot A^{-1} = A \cdot \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} \cdot A^{-1}$ is as desired.

Resolution of identity.

Def. Given a vector space V , a collection $\{E_i\}_{i=1}^k$ of linear operators on V satisfying

$$\underbrace{E_i^2 = E_i \text{ for each } 1 \leq i \leq k}_{\text{idempotent}}, \quad \underbrace{E_i \circ E_j = 0 \text{ if } i \neq j}_{\text{orthogonal}}, \quad \underbrace{I = E_1 + \dots + E_k}_{\text{complete}}.$$

is called the resolution of the identity.

In fact, the orthogonal with complete implies the idempotent, because

$$I = E_1 + \dots + E_k \Rightarrow E_i = E_i I = E_i E_1 + \dots + E_i E_k \Rightarrow E_i = E_i^2.$$

Moreover, if $\{E_i\}_{i=1}^k$ is a resolution of the identity, then

$$V = \text{Im } E_1 \oplus \dots \oplus \text{Im } E_k$$

because: given $\alpha \in V$,

$$\alpha = E_1 \alpha + \dots + E_k \alpha$$

so $V = \text{Im } E_1 + \dots + \text{Im } E_k$, that is, the complete gives the sum decomposition, and,

for each $1 \leq i \leq k$, $\alpha_i \in \text{Im } E_i$ be given. Suppose that $\alpha_1 + \dots + \alpha_k = 0$, then

$\alpha_i = E_i \beta_i$ for some $\beta_i \in V$, so, for each $1 \leq j \leq k$, $\underbrace{\text{orthogonal}}_{\text{idempotent}}$

$$0 = E_j \cdot 0 = E_j \alpha_1 + \dots + E_j \alpha_k = E_j E_1 \beta_1 + \dots + E_j E_k \beta_k = E_j E_1 \beta_1 + \dots + E_j E_j \beta_j = E_j \beta_j = \alpha_j.$$

Thus $\text{Im } E_i$ are independent, so proved.

Lemma. Let T be a linear operator on V , and $E_1, \dots, E_k$ be the projections satisfying

$$E_i \circ E_j = 0 \text{ if } i \neq j \text{ and } I = \sum_{i=1}^k E_i.$$

Then,

For each $\text{Im } E_i$ is invariant under T if and only if $TE_i = E_i T$ for each $1 \leq i \leq k$.

Proof. Suppose that $\text{Im } E_1, \dots, \text{Im } E_k$ are invariant under T .

Given $\alpha \in V$, $\alpha = E_1 \alpha + \dots + E_k \alpha$ and so $T\alpha = TE_1 \alpha + \dots + TE_k \alpha$.

By the assumption, each $TE_i \alpha \in \text{Im } E_i$, so $TE_i \alpha = E_i \beta_i$ for some $\beta_i \in V$.

Therefore,

$$E_i T \alpha = E_i (TE_1 \alpha + \dots + TE_k \alpha) = E_i (E_1 \beta_1 + \dots + E_k \beta_k) = E_i E_i \beta_i = E_i \beta_i = TE_i \alpha.$$

So commute. Conversely, let $\alpha \in \text{Im } E_i$. Then, $T\alpha = TE_i \alpha = E_i T \alpha \in \text{Im } E_i$, so proved. $\square$

Thm. Let T be a diagonalizable linear operator on a finite-dimensional space V , with distinct eigenvalues $c_1, \dots, c_k \in F$.

Then, there exists a resolution of identity $\{E_i\}_{i=1}^k$ on V satisfying

$$T = c_1 E_1 + \dots + c_k E_k \quad \text{and for each } 1 \leq i \leq k, \operatorname{Im} E_i = \ker(T - c_i I).$$

Proof. By the equivalent condition of diagonalizable, we obtain directly that

$$V = \ker(T - c_1 I) \oplus \dots \oplus \ker(T - c_k I)$$

Now, for each $1 \leq i \leq k$, let E_i be the projection on $\ker(T - c_i I)$ along $\bigoplus_{j \neq i} \ker(T - c_j I)$.

Polynomial evaluation of linear operation.

Let T be a diagonalizable operation on a fin. dim space V with a resolution

$$T = c_1 E_1 + \dots + c_k E_k.$$

Then, the orthogonality gives that: for any polynomial $g(t) \in F[t]$,

$$g(T) = g(c_1) E_1 + \dots + g(c_k) E_k.$$

So, we can use the Lagrange Polynomial,

$$p_j(t) = \prod_{i \neq j} \frac{(t - c_i)}{(c_j - c_i)}, \text{ so that } p_j(T) = E_j.$$

Thm. Let V be a fin. dim. v.s. over F .

If $V = \bigoplus_{i=1}^k W_i$, then there exist k linear operators E_i on V s.t.

$$E_i \circ E_j = \delta_{ij} \cdot E_i, \quad I = \sum_{i=1}^k E_i, \quad W_i = \text{Im } E_i \quad (1 \leq i \leq k) \quad \dots (*)$$

Conversely, if $E_1, \dots, E_k$ are k linear operators on V which satisfy $(*)$, then

$$V = \bigoplus_{i=1}^k W_i$$

Proof. Suppose $V = \bigoplus_{i=1}^k W_i$. For each $1 \leq i \leq k$, define $E_i: V \rightarrow V: \sum_{j=1}^k \alpha_j v_j \mapsto \alpha_i v_i$. Proved.

Conversely, suppose that $E_1, \dots, E_k$ are linear operators on V satisfying $(*)$.

Given $\alpha \in V$, $\alpha = E_1(\alpha) + \dots + E_k(\alpha) \in W_1 + \dots + W_k$. And, to show uniqueness such expression,

let $\alpha = \alpha_1 + \dots + \alpha_n$, where $\alpha_i \in W_i$. And, let $\alpha_i = E_i(\beta_i)$, for each $1 \leq i \leq k$.

Now, for each j , $1 \leq j \leq k$,

$$E_j(\alpha) = \sum_{i=1}^k E_j(\alpha_i) = \sum_{i=1}^k (E_j \circ E_i)(\beta_i) = E_j^2(\beta_j) = E_j(\beta_j) = \alpha_j$$

Thus Proved.

Layer 2

Lemma. Let T be a linear operator on V , and $E_1, \dots, E_k$ be the projections satisfying

$$E_i \circ E_j = 0 \text{ if } i \neq j \text{ and } I = \sum_{i=1}^k E_i.$$

Then,

For each $\text{Im } E_i$ is invariant under T if and only if $TE_i = E_i T$ for each $1 \leq i \leq k$.

Proof. Suppose that $\text{Im } E_1, \dots, \text{Im } E_k$ are invariant under T .

Given $\alpha \in V$, $\alpha = E_1 \alpha + \dots + E_k \alpha$ and so $T\alpha = TE_1 \alpha + \dots + TE_k \alpha$.

By the assumption, each $TE_i \alpha \in \text{Im } E_i$, so $TE_i \alpha = E_i \beta_i$ for some $\beta_i \in V$.

Therefore,

$$E_i T \alpha = E_i (TE_1 \alpha + \dots + TE_k \alpha) = E_i (E_i \beta_1 + \dots + E_k \beta_k) = E_i E_i \beta_i = E_i \beta_i = TE_i \alpha.$$

So converse. Conversely, let $\alpha \in \text{Im } E_i$. Then, $T\alpha = TE_i \alpha = E_i T \alpha \in \text{Im } E_i$, so proved. $\square$